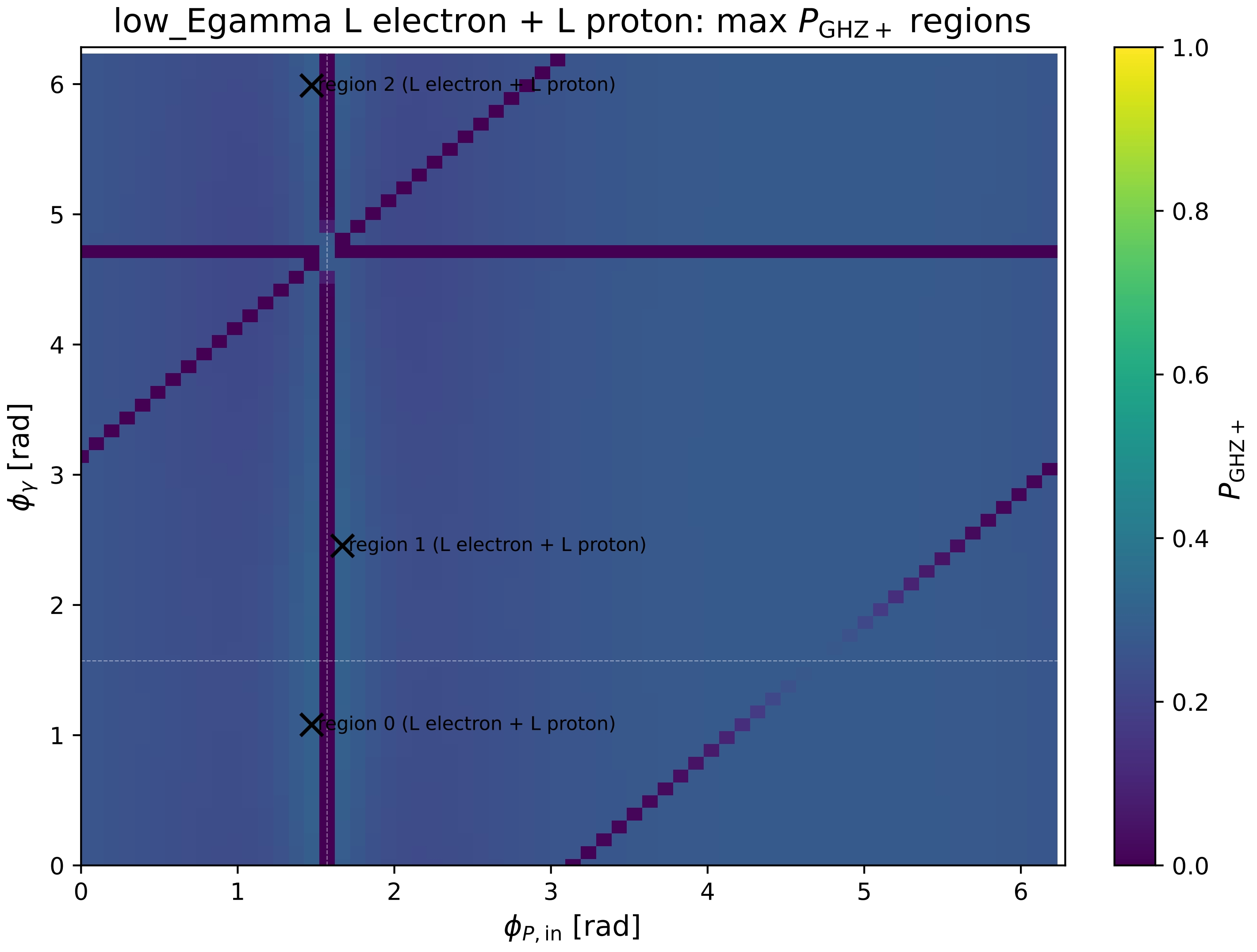
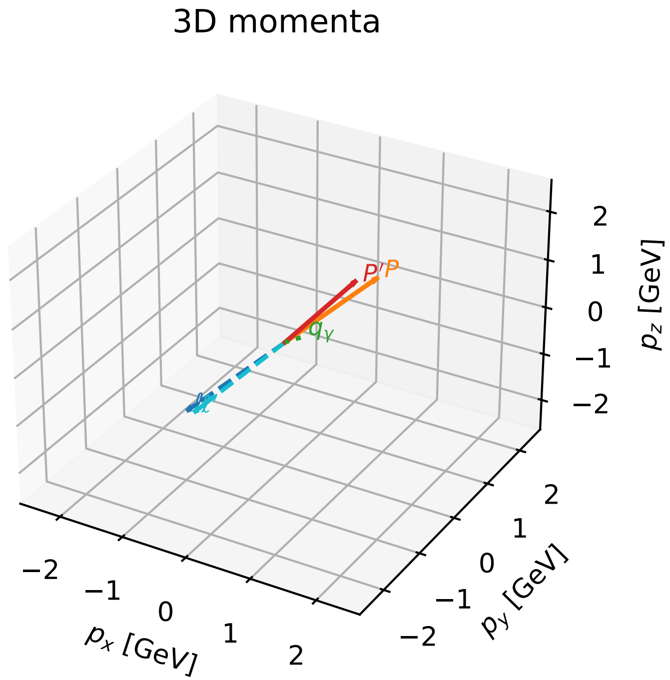


low_Egamma L electron + L proton: max $P_{\mathrm{GHZ}+}$ regions
region 2 (L electron + L proton)
region 1 (L electron + L proton)
region 0 (L electron + L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

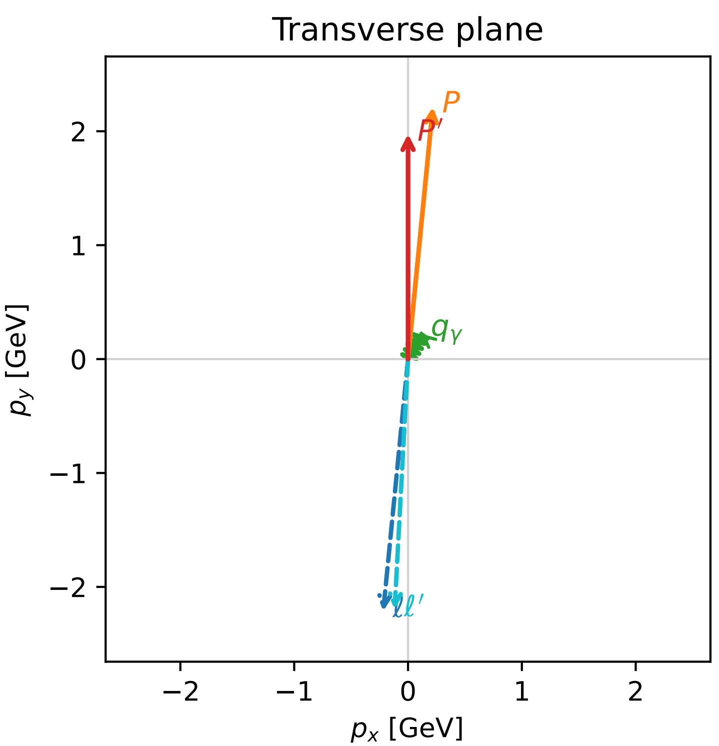
L electron + L proton characteristic max $P_{\mathrm{GHZ+}}$ configuration

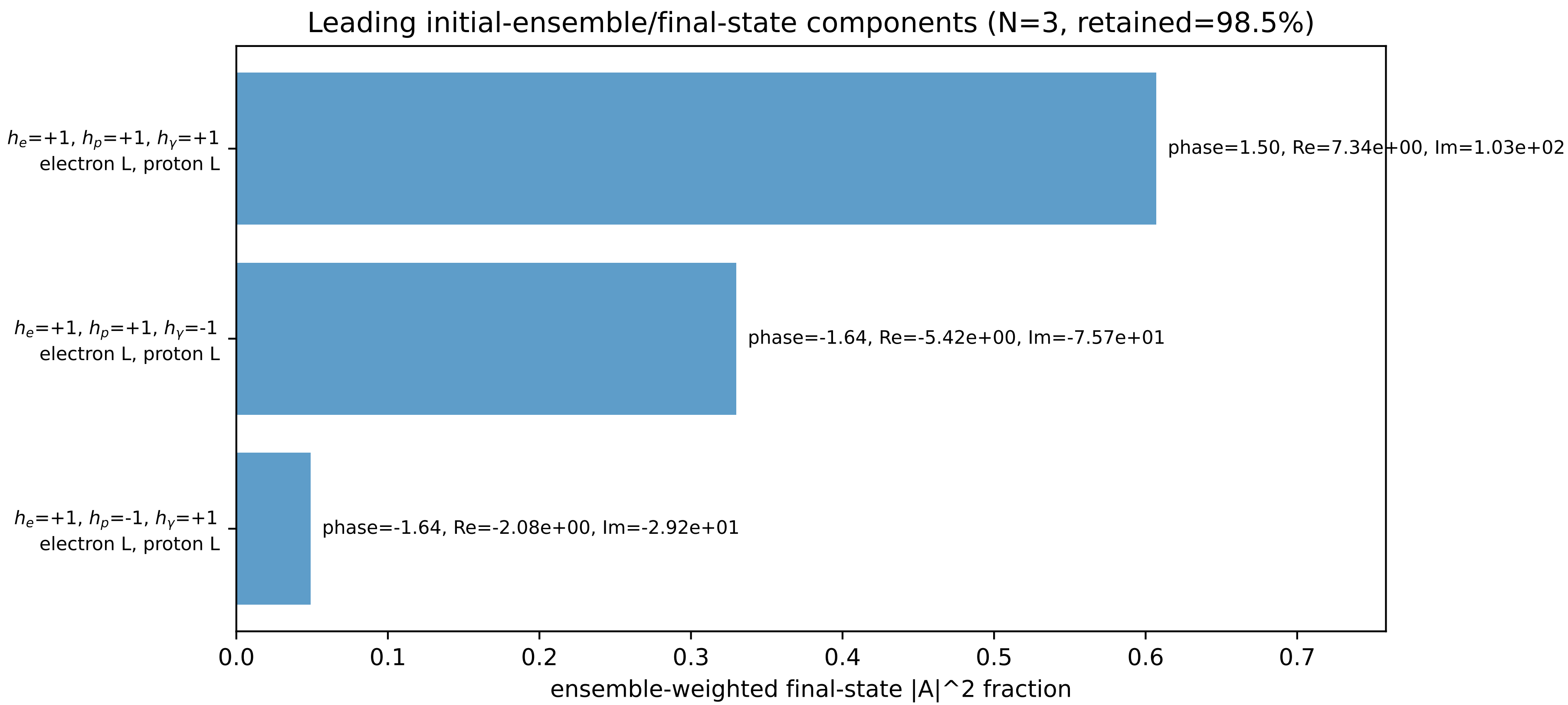


ghz_purity_low_egamma_ll_region_0 P_{\rm GHZ+}=0.303429
region: low_Egamma_LL_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$

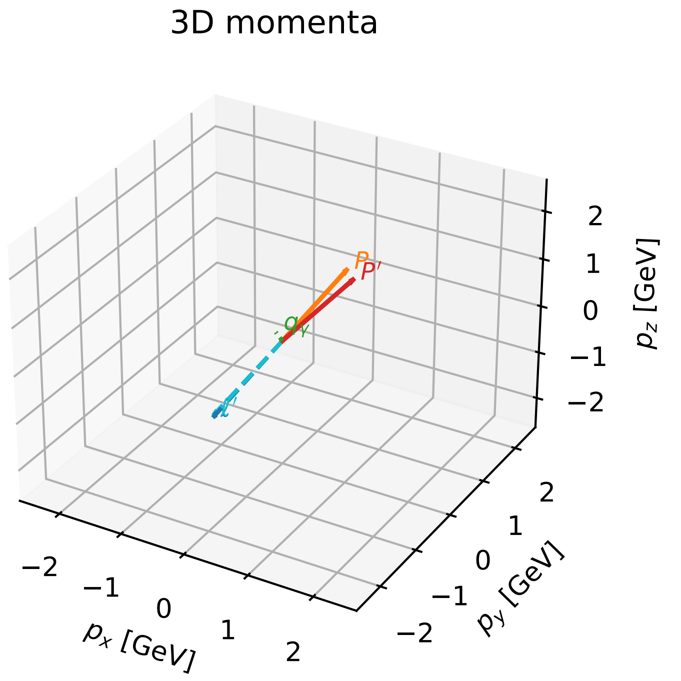
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.97681
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.61421, ϕ_P=1.47262
E_γ=0.25, ϕ_γ=1.07992
Q^2=0.00973142, x_B=0.0419335, t=-0.0525649, W^2=1.10218, y=0.0112458
$\theta(\ell', \gamma)$=154.945 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.2180 p_y= -2.2137 p_z= -0.0000
P E= 2.4141 p_x= 0.2180 p_y= 2.2137 p_z= 0.0000
ℓ' E= 2.2005 p_x= -0.1178 p_y= -2.1973 p_z= 0.0000
P' E= 2.1881 p_x= 0.0000 p_y= 1.9768 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1178 p_y= 0.2205 p_z= 0.0000





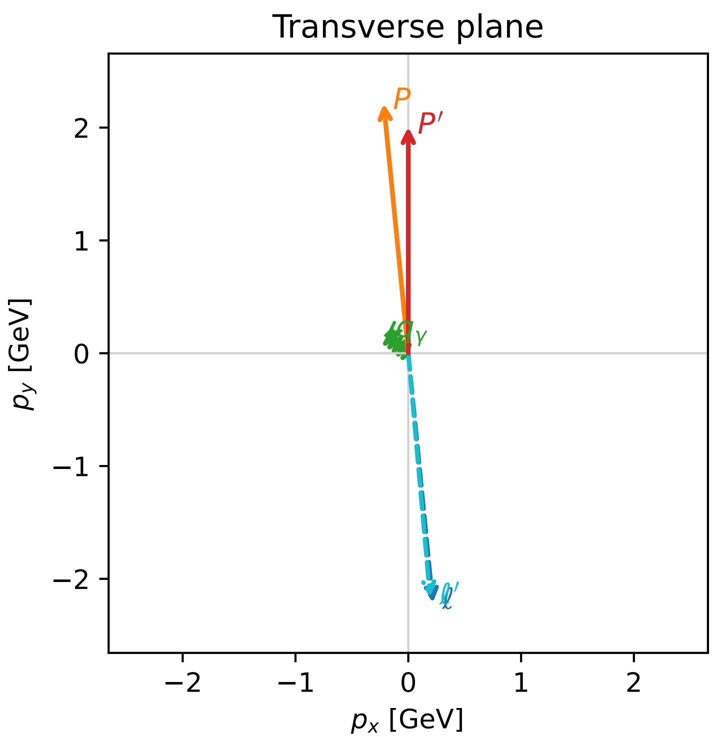
L electron + L proton characteristic max $P_{\mathrm{GHZ}+}$ configuration

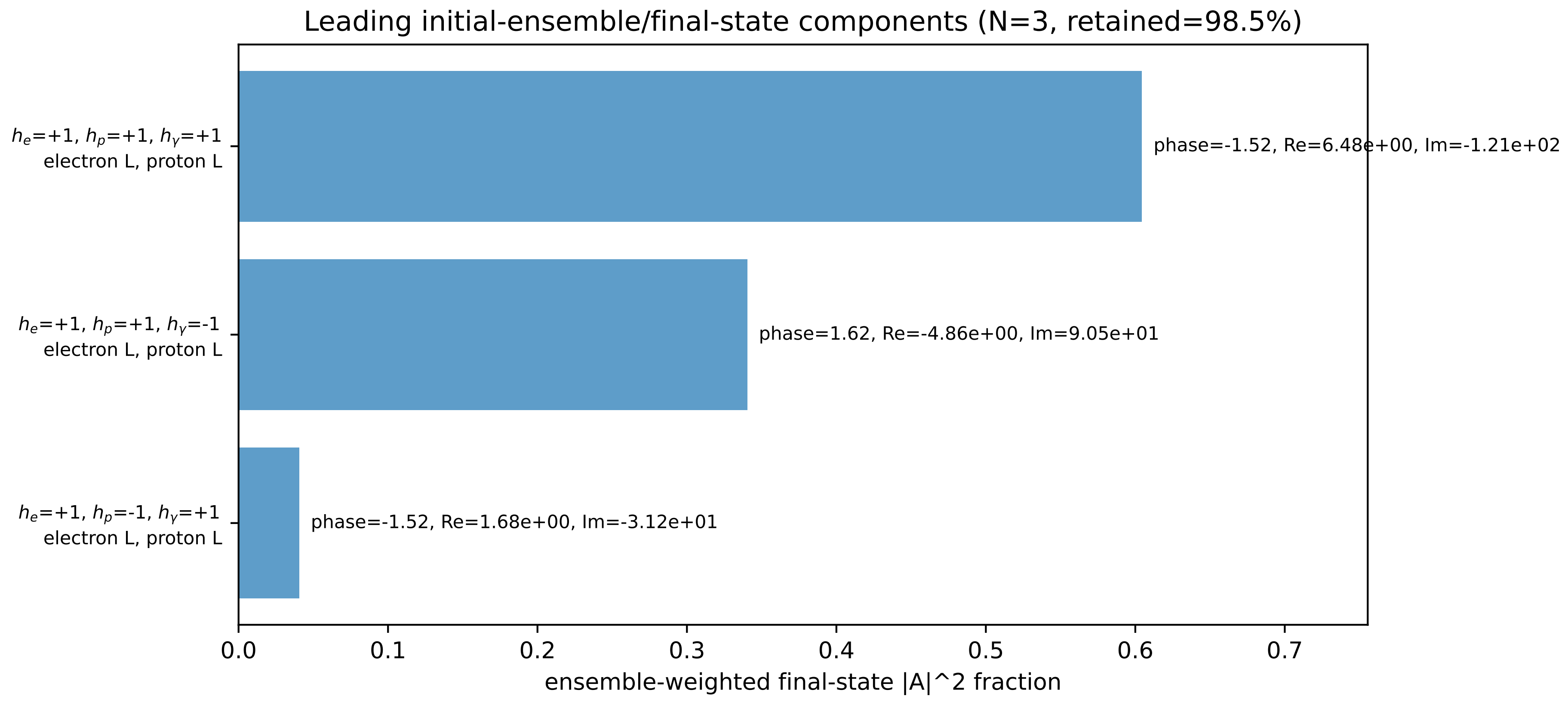


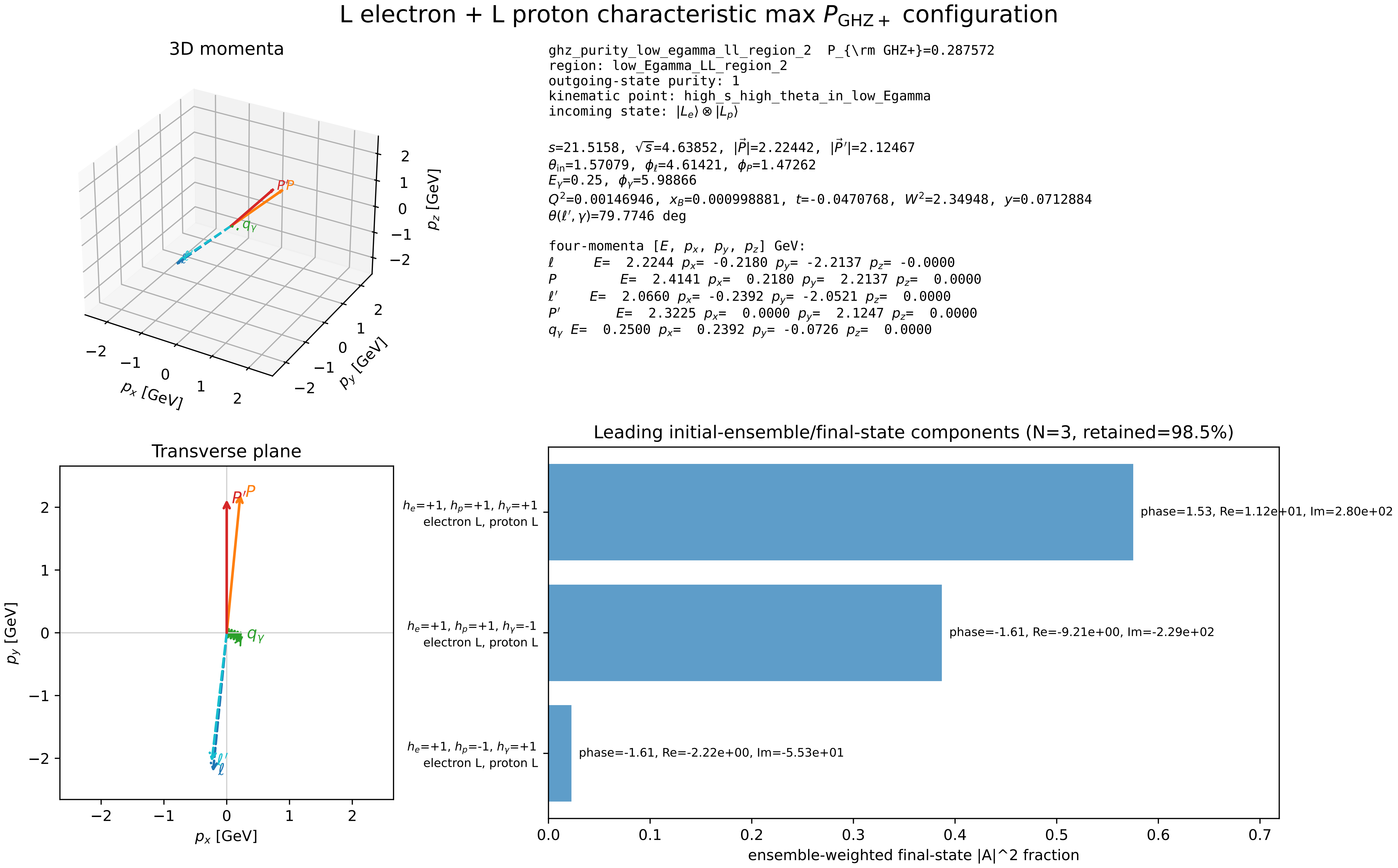
ghz_purity_low_egamma_ll_region_1 P_{\rm GHZ+}=0.302205
region: low_Egamma_LL_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.00644
θ_{in}=1.57079, ϕ_ℓ=4.81056, ϕ_P=1.66897
E_γ=0.25, ϕ_γ=2.45437
Q^2=0.000404817, x_B=0.000858749, t=-0.050804, W^2=1.35084, y=0.0228437
$\theta(\ell', \gamma)$=134.476 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.2180 p_y= -2.2137 p_z= -0.0000
P E= 2.4141 p_x= -0.2180 p_y= 2.2137 p_z= 0.0000
ℓ' E= 2.1736 p_x= 0.1933 p_y= -2.1650 p_z= 0.0000
P' E= 2.2149 p_x= 0.0000 p_y= 2.0064 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1933 p_y= 0.1586 p_z= 0.0000






L electron + L proton characteristic max $P_{\mathrm{GHZ+}}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P' P
q_γ
ℓ' ℓ
ghz_purity_low_egamma_ll_region_2 P_{\rm GHZ+}=0.287572
region: low_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.12467
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.61421, ϕ_P=1.47262
E_γ=0.25, ϕ_γ=5.98866
Q^2=0.00146946, x_B=0.000998881, t=-0.0470768, W^2=2.34948, y=0.0712884
$\theta(\ell', \gamma)$=79.7746 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.2180 p_y= -2.2137 p_z= -0.0000
P E= 2.4141 p_x= 0.2180 p_y= 2.2137 p_z= 0.0000
ℓ' E= 2.0660 p_x= -0.2392 p_y= -2.0521 p_z= 0.0000
P' E= 2.3225 p_x= 0.0000 p_y= 2.1247 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2392 p_y= -0.0726 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=3, retained=98.5%)
h_e=+1, h_p=+1, h_γ=+1 electron L, proton L
phase=1.53, Re=1.12e+01, Im=2.80e+02
h_e=+1, h_p=+1, h_γ=-1 electron L, proton L
phase=-1.61, Re=-9.21e+00, Im=-2.29e+02
h_e=+1, h_p=-1, h_γ=+1 electron L, proton L
phase=-1.61, Re=-2.22e+00, Im=-5.53e+01
ensemble-weighted final-state |A|^2 fraction